

GxP COMPUTERIZED SYSTEM LIFECYCLE

GxP LIMS Lifecycle Documentation Package

Laboratory Information Management System (LIMS)

Document ID	LIMS-LCP-001
Version	0.1
Status	DRAFT FOR REVIEW - NOT A CONTROLLED GxP DOCUMENT

Note: this document list for LIMS system is shared for Hackathon event, this are not actual list. Shared for your learning purpose.



Figure 1. Risk-based lifecycle roadmap used for the LIMS documentation package.

Purpose

The purpose is to provide a complete document inventory and control model that enables a project and operations team to plan, create, review, approve, maintain and retire the evidence required to demonstrate that LIMS is fit for intended use, that GxP data remain reliable throughout their lifecycle, and that the system remains in a validated state.

Complete Lifecycle Document Inventory

Mandatory (M) items are expected for the baseline LIMS scope. Conditional (C) items become mandatory when the stated feature, hosting model, migration, custom development, personal-data processing, AI/ML use or other triggering condition applies. Documents may be combined when purpose, traceability, approval and readability are preserved.

1 - Concept and Initiation

ID	Document / record	M/C	Purpose	Owner	Reviewer(s)	Approver(s)	Regulatory / SOP basis	Inputs -> outputs
1.01	Business Need and Benefits Case	M	Defines the quality/business problem, intended value and why LIMS is required.	Business Process Owner	Finance; QC leadership; IT	Business sponsor	ICH Q10; Q187219 sections 1 and 4	Business strategy -> approved need and funding
1.02	Intended Use and Scope Statement	M	Defines supported laboratory processes, sites, users, records, exclusions and boundary conditions.	Business Process Owner	IT System Manager; QA; Data Owner	Business Process Owner; QA	Annex 11 Principle/4.4; 21 CFR 11.10(a); Q187219 1/4.3	Process description -> approved intended use
1.03	GxP Criticality and System Classification Assessment	M	Classifies LIMS as a GxP-critical IT system and records impacts to product quality, patient safety and GxP data.	IT System Manager	Business Process Owner; QA	IT System Owner; QA	Q187219 section 1, Table 3	Intended use -> criticality classification
1.04	Software Category Assessment	M	Classifies software components (typically category 4 configured application, with category 5 elements if custom code exists).	IT System Manager	Supplier SME; System Architect; QA	IT System Owner; QA	GAMP 5; Q187219 Appendix 2	Architecture and supplier information -> category rationale
1.05	Electronic Records and Electronic Signatures Applicability Assessment	M	Determines Part 11 and Annex 11 electronic record/signature applicability and predicate-rule records.	QA / Validation Lead	Business Process Owner; Data Owner; Legal/Regulatory	QA	21 CFR 11.1/11.10/11.50/11.70; Annex 11 section 14	Record inventory -> applicable control set
1.06	System Registration and Inventory Record	M	Registers the LIMS business application, owner, criticality, lifecycle status and service dependencies.	IT System Manager	IT System Owner	Per supporting IT solution	Annex 11 4.3; Q187219 record IT system	Classification -> ServiceNow ITOM record
1.07	Project Charter and Governance Plan	M	Defines objectives, scope, decision rights, milestones, governance forums and escalation routes.	Project Manager / Product Owner	Business; IT; QA; Supplier	Steering committee	ICH Q10; Q187219 4.3	Approved need -> governed implementation
1.08	Role Appointment and Delegation Record	M	Appoints IT System Owner, IT System Manager, Business Process Owner, Data Owner, Data Responsible, QA and technical SMEs.	IT System Owner	QA; Business Process Owner	IT System Owner	Annex 11 section 2; Q187219 roles	Governance plan -> accountable roles
1.09	Initial Process and Data Flow Map	M	Maps high-level laboratory workflow and movement of GxP records through instruments, interfaces, LIMS and archives.	Business Process Owner / Data Owner	IT System Manager; Lab SMEs; Supplier	Business Process Owner	Annex 11 4.3/5; Q187219 2/4.5	Process knowledge -> initial flow map
1.10	Initial IT Risk Assessment (ITRA) Screening	M	Identifies initial information-security, data-integrity, privacy and continuity risks to support planning and sourcing.	IT System Manager	QA; Security; Data Owner; Privacy	IT System Owner; QA	Q187219 section 2; Q0300378 section 1	Intended use and flow -> draft risks and controls
1.11	Supplier Sourcing and Evaluation Strategy	M	Defines supplier due diligence, audit need, QMS reliance and contracting approach.	Procurement / IT System Manager	QA; Security; Business; Legal	IT System Owner; Procurement	Annex 11 section 3; Q187219 4.1	Need and risks -> sourcing/evaluation plan

ID	Document / record	M/C	Purpose	Owner	Reviewer(s)	Approver(s)	Regulatory / SOP basis	Inputs -> outputs
1.12	Initial Supplier Assessment Conclusion	M	Assesses supplier competence, reliability, QMS, development, verification, security, data handling and continuity.	IT System Manager	QA; Security; Privacy; Procurement	Peer; QA quality aspects	Annex 11 3.2/4.5; Q187219 4.1	Supplier evidence -> assessment and mitigations
1.13	Initial Change Request / Quality Change	M	Authorises implementation and ensures regulatory/quality impact is evaluated under change control.	Change Owner	Business; IT; QA	Change Control Board / QA	Annex 11 10; Q187219 4.2	Project scope -> approved change record
1.14	Validation / Implementation Plan (VPL)	M	Defines lifecycle strategy, scope, deliverables, responsibilities, supplier reliance, infrastructure qualification, verification and release.	Validation Lead / IT System Manager	Business; Technical SMEs; Supplier	IT System Owner; QA signature	Annex 11 4.1; Q187219 4.3 and Appendix 1	Classification and risks -> approved lifecycle plan
1.15	Documentation and Records Management Plan	M	Defines repositories, identifiers, version control, approval, retention, traceability and supplier-document acceptance.	Validation Lead	Document Control; QA; IT	QA	21 CFR 11.10(k); Annex 11 4.1; Q187219 4.3	VPL -> controlled documentation approach
1.16	Quality Oversight Plan	C	Defines QA presence, surveillance and risk-based oversight for complex, global or supplier-hosted implementation.	QA	IT System Manager; Business Process Owner	QA management	ICH Q10; Q187219 roles	Project risk -> QA oversight activities
1.17	Organisational Change and Training Strategy	M	Plans role changes, SOPs, curricula, training qualification and go-live readiness.	Business Process Owner	Training; QA; IT	Business Process Owner; QA	21 CFR 11.10(i); Q187219 4.3/4.8	Process impact -> training and adoption plan

17 lifecycle documents/records in this phase

2 - Requirements

ID	Document / record	M/C	Purpose	Owner	Reviewer(s)	Approver(s)	Regulatory / SOP basis	Inputs -> outputs
2.01	User Requirements Specification (URS)	M	Defines complete, correct, unambiguous and verifiable functional and non-functional needs.	Business Process Owner	Lab SMEs; Data Owner; IT; Supplier	IT System Owner; QA	Annex 11 4.4; Q187219 4.4	Intended use and risks -> approved requirements
2.02	GxP Requirement and Critical Aspect Classification	M	Identifies GxP user requirements and critical aspects whose failure may affect product quality or patient safety.	Validation Lead / QA	Business; Lab SMEs; IT	QA	Q187219 2/4.4; ICH Q9(R1)	URS -> risk-ranked requirements
2.03	Data Requirements Specification	M	Defines required GxP records, metadata, ownership, source, lineage, review, retention, retrieval, reporting and disposition.	Data Owner	Business; QA; IT; Records Management	Data Owner; QA	Annex 11 5/7/8/9/17; PIC/S PI 041-1	Data inventory -> approved data requirements
2.04	Data Integrity and ALCOA+ Requirements	M	Translates attributable, legible, contemporaneous, original, accurate, complete, consistent, enduring and available principles into testable requirements.	Data Owner / QA	Business; IT; Supplier	QA	FDA DI guidance; PIC/S PI 041-1; Q204010 reference	Risk assessment -> DI requirements
2.05	Information Security Requirements	M	Defines confidentiality, integrity, availability, logging, vulnerability, patching, encryption, remote-access and monitoring requirements.	Information Security SME	IT; Supplier; Data Owner; QA	Security authority / IT System Owner	Annex 11 12; Q187655 reference	Threats and architecture -> security requirements

ID	Document / record	M/C	Purpose	Owner	Reviewer(s)	Approver(s)	Regulatory / SOP basis	Inputs -> outputs
2.06	User Access and Segregation of Duties Requirements	M	Defines roles, least privilege, privileged access, service accounts, approvals, revocation and periodic review.	Business Process Owner / IT	QA; Security; Data Owner	Business Process Owner; QA	Annex 11 12; 21 CFR 11.10(d)/(g)	Process roles -> access model requirements
2.07	Audit Trail Requirements Specification	M	Defines events to audit, content, time source, security, retention, availability, review and reporting.	QA / Data Owner	Business; IT; Supplier	QA	Annex 11 9; 21 CFR 11.10(e)	Critical data actions -> audit trail requirements
2.08	Electronic Signature Requirements Specification	C	Defines signature meaning, identity, manifestation, linkage, uniqueness, controls and accountability if electronic signatures are used.	QA / Business Process Owner	IT; Legal; Supplier	QA	Annex 11 14; 21 CFR 11.50/11.70/11.100-11.300	Signature assessment -> e-signature requirements
2.09	Reporting, True Copy and Export Requirements	M	Defines human-readable and electronic copies, report content, version/status indication, changed-data visibility and export integrity.	Business Process Owner / Data Owner	QA; IT; Supplier	Business Process Owner; QA	Annex 11 8; 21 CFR 11.10(b)	Record use cases -> report/copy requirements
2.10	Interface and Data Exchange Requirements	M	Defines interface message content, security, acknowledgements, error handling, reconciliation, retries, duplicate prevention and monitoring.	System Architect / Data Owner	Business; QA; Supplier SMEs	IT System Owner; QA	Annex 11 5; Q187219 4.5	Data flow -> interface requirements
2.11	Master and Reference Data Requirements	M	Defines ownership, lifecycle, dual review, effective dating, versioning, controlled changes and impact assessment.	Data Owner / Lab Process Owner	QA; IT; Supplier	Data Owner; QA	PIC/S PI 041-1; Annex 11 5/10	Process rules -> master data requirements
2.12	Calculation and Scientific Logic Requirements	M	Defines formulas, units, rounding, significant figures, limits, specifications, result flags and exception handling.	Lab Process Owner / Method SME	QA; Data Owner; Supplier	Business Process Owner; QA	Annex 11 4.7/6; data integrity guidance	Methods/specifications -> calculation requirements
2.13	Backup, Restore and Recovery Requirements	M	Defines backed-up objects, frequency, retention, RPO, RTO, WRT, MTD, restore integrity and recovery order.	IT System Manager / Business Continuity Owner	Business; Data Owner; Infrastructure; QA	IT System Owner; Business Process Owner	Annex 11 7.2/16; Q187219 4.11	BIA and data criticality -> recovery requirements
2.14	Retention, Archive and Retrieval Requirements	M	Defines retention periods, legal holds, archive format, readability, metadata, access, retrieval time and integrity verification.	Data Owner / Records Management	QA; Legal; IT	Data Owner; QA	Annex 11 7/17; 21 CFR 11.10(c)	Record inventory -> retention/archive requirements
2.15	Performance, Capacity and Availability Requirements	M	Defines loads, response times, throughput, concurrency, growth, uptime and monitoring thresholds.	Business Process Owner / Architect	IT Operations; Supplier; QA	IT System Owner	Annex 11 Principle/16; Q187219 4.4	Usage forecast -> measurable NFRs
2.16	Business Continuity and Manual Fallback Requirements	M	Defines alternative workflows, data capture, reconciliation, activation time, responsibilities and testing.	Business Continuity Owner	Lab Operations; QA; IT	Business Process Owner; QA	Annex 11 16	BIA -> fallback requirements
2.17	Data Migration and Conversion Requirements	C	Defines scope, mapping, transformation, cleansing, reconciliation, exception handling, lineage and approval for legacy data.	Data Owner / Migration Lead	Business; QA; IT; Supplier	Data Owner; QA	Annex 11 4.8	Legacy data scope -> migration requirements
2.18	Infrastructure and Hosting Requirements	M	Defines qualified infrastructure, environments, network, database, time sync, monitoring, backup, security and support prerequisites.	System Architect / Infrastructure Owner	Security; Supplier; QA	IT Infrastructure Owner / QA as applicable	Annex 11 Principle; Q216301	Application needs -> infrastructure requirements

ID	Document / record	M/C	Purpose	Owner	Reviewer(s)	Approver(s)	Regulatory / SOP basis	Inputs -> outputs
2.19	Privacy and Personal Data Requirements	C	Defines lawful processing, minimisation, retention, subject rights, access and breach controls if personal data beyond user identities are processed.	Business Activity Owner / Privacy	Data Owner; Security; Legal	Privacy authority	Q187219 section 2; applicable privacy law	Personal data assessment -> privacy requirements
2.20	AI/ML Applicability and Requirements Assessment	C	Confirms AI/ML is excluded or defines intended use, model/data governance, performance, change, monitoring and human oversight.	Business Process Owner / AI Owner	QA; Data Owner; Security; Legal	QA; Governance body	Internal AI governance; GAMP risk principles	Feature scope -> AI/ML control decision

20 lifecycle documents/records in this phase

3 - Risk Management

ID	Document / record	M/C	Purpose	Owner	Reviewer(s)	Approver(s)	Regulatory / SOP basis	Inputs -> outputs
3.01	Approved IT Risk Assessment (ITRA)	M	Assesses information security, GxP data integrity, privacy and technology risks; defines risk and lifecycle controls.	IT System Manager	QA; Security; Data Owner; Business	IT System Owner residual risk; QA	Q187219 section 2	Risks and requirements -> approved controls
3.02	Functional Risk Assessment / ITRRA	M	Assesses failure modes of GxP requirements and functions, identifies critical aspects and verification depth.	Validation Lead	Business SMEs; IT; Supplier	Peer; QA	Q187219 section 2/4.4	URS -> risk-ranked functions and tests
3.03	Data Integrity Risk Assessment	M	Assesses data lifecycle risks from creation through disposal, including metadata, audit trails, interfaces, manual entries and administration.	Data Owner / QA	Business; IT; Supplier	Data Owner; QA	FDA DI guidance; PIC/S PI 041-1; Annex 11	Data flow/inventory -> DI controls
3.04	Cybersecurity Threat and Vulnerability Assessment	M	Assesses threats, vulnerabilities, attack paths, remote access, third parties and compensating controls.	Information Security SME	Architect; Supplier; IT Operations	Security authority	Annex 11 12; Q187655 reference	Architecture -> security controls and residual risks
3.05	Supplier Risk Assessment and Audit Decision	M	Assesses supplier QMS, product/service risk, hosting, subcontractors, continuity and need for audit.	IT System Manager / Procurement	QA; Security; Privacy; Business	Peer; QA quality aspects	Annex 11 3; Q187219 4.1	Supplier evidence -> controls and audit decision
3.06	Business Continuity Impact Assessment	M	Determines critical business processes, outage impacts, MTD, RTO, WRT, RPO and fallback priorities.	Business Process Owner	IT; QA; Data Owner	Business Process Owner	Annex 11 16; Q187219 4.11	Process criticality -> recovery objectives
3.07	Interface Risk Assessment	M	Assesses loss, duplication, truncation, alteration, delay, mapping and reconciliation risks for each interface.	Data Owner / Interface SME	Business; QA; IT	Data Owner; QA	Annex 11 5	Interface inventory -> control design
3.08	Data Migration Risk Assessment	C	Assesses legacy-data quality, mapping, transformation, volume, reversibility, reconciliation and archive risks.	Migration Lead / Data Owner	QA; Business; IT; Supplier	Data Owner; QA	Annex 11 4.8	Migration requirements -> migration controls
3.09	Electronic Signature Risk Assessment	C	Assesses identity, authentication, non-repudiation, signature meaning, linkage and credential risks.	QA / Security	Business; Legal; IT	QA	21 CFR Part 11; Annex 11 14	Signature requirements -> control design

ID	Document / record	M/C	Purpose	Owner	Reviewer(s)	Approver(s)	Regulatory / SOP basis	Inputs -> outputs
3.10	Privacy Risk Assessment / DPIA	C	Assesses risk to individuals and identifies data-protection controls where required.	Business Activity Owner	Privacy; Security; Data Owner	Privacy authority	Q187219 section 2; applicable privacy law	Processing description -> privacy controls
3.11	Risk Control Implementation Tracker	M	Tracks each risk control to specification, configuration, procedure, evidence, owner and status.	Validation Lead / IT System Manager	Risk owners; QA	QA at lifecycle gates	ICH Q9(R1); Q187219 section 2	Risk assessments -> implemented/verified controls
3.12	Residual Risk Acceptance Record	M	Documents remaining risks, rationale, ownership, monitoring and formal acceptance before release.	Risk Owner / IT System Owner	Business; QA; Security; Data Owner	IT System Owner; QA where quality risk	ICH Q9(R1); Q187219 Appendix 1	Verified controls -> accepted residual risk

12 lifecycle documents/records in this phase

4 - Design

ID	Document / record	M/C	Purpose	Owner	Reviewer(s)	Approver(s)	Regulatory / SOP basis	Inputs -> outputs
4.01	System Description	M	Provides inspector-ready overview of supported processes, functionality, physical/logical architecture, data flows, interfaces, prerequisites and security.	IT System Manager	Architect; Data Owner; Business; QA	Per QMS / acceptance	Annex 11 4.3; Q187219 4.8	Approved design -> maintained system description
4.02	Solution Architecture and Environment Diagram	M	Defines application, database, middleware, client, network, identity, monitoring, backup and environment topology.	System Architect	Infrastructure; Security; Supplier	Architecture authority	Annex 11 4.3; Q187219 4.5	Requirements -> architecture baseline
4.03	Functional Specification (FS)	M	Defines system behaviour required to fulfil URS, including workflows, calculations, controls and exceptions.	Supplier / Functional Lead	Business SMEs; Validation; QA	Peer / designated approver	Q187219 4.5; GAMP 5	URS and risks -> functional design
4.04	Design / Technical Specification (DS/TDS)	C	Defines technical implementation for custom, complex or integration components.	Supplier / Technical Lead	Architect; Security; Validation	Peer	Q187219 4.5; GAMP category 5 where applicable	FS -> technical design
4.05	Configuration Specification (CS)	M	Defines configured workflows, fields, states, rules, parameters, calculations, reports, roles and settings.	Configuration Lead	Business SMEs; Validation; QA	Peer / designated approver	Q187219 4.5; GAMP category 4	URS/FS -> configuration baseline
4.06	Data Model and Metadata Specification	M	Defines entities, identifiers, relationships, metadata, status, versioning, auditability and retention attributes.	Data Architect / Supplier	Data Owner; QA; IT	Data Owner / Architecture authority	Annex 11 4.3/5/7/9	Data requirements -> governed data model
4.07	Detailed Data Flow Diagram	M	Shows each source, transformation, storage, review, output, archive and control boundary.	Data Owner / Architect	Business; QA; Interface SMEs	Data Owner; QA	Annex 11 4.3/5	Process/data requirements -> controlled flow
4.08	Interface Design Specification / Terms of Service	M	Defines message schemas, mapping, security, timing, acknowledgements, retries, error queues, reconciliation and ownership.	Interface Lead	Data Owner; Business; QA; Supplier	Architecture authority; QA	Annex 11 5; Q187219 4.5	Interface requirements -> interface design

ID	Document / record	M/C	Purpose	Owner	Reviewer(s)	Approver(s)	Regulatory / SOP basis	Inputs -> outputs
4.09	Security Design and Hardening Specification	M	Defines authentication, authorisation, encryption, logging, network controls, hardening, vulnerability and remote-access design.	Security Architect	IT Operations; Supplier; QA	Security authority	Annex 11 12; 21 CFR 11.10(d)/(g)	Security requirements -> security design
4.10	Role, Permission and Segregation of Duties Design	M	Maps business roles to functions, data scope, approval authority, privileged activities and conflicting access.	Business Process Owner / IT	QA; Security; Data Owner	Business Process Owner; QA	Annex 11 12; 21 CFR 11.10(g)	Access requirements -> role design
4.11	Audit Trail Design Specification	M	Defines audited events, old/new values, identity, time, reason, review views, retention and export.	Supplier / IT	QA; Data Owner; Business	QA	Annex 11 9; 21 CFR 11.10(e)	Audit trail requirements -> technical design
4.12	Electronic Signature Design Specification	C	Defines signature points, meaning, authentication, manifestation, linkage, reauthentication and controls.	Supplier / IT	QA; Business; Security	QA	Annex 11 14; 21 CFR 11.50/11.70	Signature requirements -> signature design
4.13	Backup, Restore and IT Recovery Design	M	Defines protected objects, schedules, retention, encryption, immutability, restore sequence, dependencies and recovery procedures.	Infrastructure / Supplier	IT System Manager; Data Owner; Business; QA	Infrastructure Owner / IT System Owner	Annex 11 7.2/16; Q187219 4.11	Recovery requirements -> recovery design
4.14	Archive and Retrieval Design	M	Defines archive storage, format, metadata, access, readability, integrity checks, search, retrieval and migration approach.	Data Owner / Architect	Records Management; QA; IT	Data Owner; QA	Annex 11 17; 21 CFR 11.10(c)	Retention requirements -> archive design
4.15	Master Data and Specification Design	M	Defines controlled creation, review, effective dates, status, versioning, impact assessment and change propagation.	Lab Process Owner / Data Owner	QA; Supplier; IT	Data Owner; QA	PIC/S PI 041-1; Annex 11 10	Master data requirements -> configuration design
4.16	Report and Certificate Design Specification	M	Defines content, source fields, calculations, version/status, signatures, changed-data visibility and true-copy controls.	Business Process Owner / Data Owner	QA; Supplier	Business Process Owner; QA	Annex 11 8; 21 CFR 11.10(b)	Reporting requirements -> report design
4.17	Data Migration Design and Mapping Specification	C	Defines source-target mapping, transformations, cleansing, defaults, exceptions, reconciliation and rollback.	Migration Lead	Data Owner; Business; QA; Supplier	Data Owner; QA	Annex 11 4.8	Migration requirements -> executable mapping
4.18	Infrastructure Design and Qualification Reference	M	Documents or references qualified infrastructure and evidence supporting the LIMS operating environment.	Infrastructure Owner	System Architect; QA	QA as applicable	Annex 11 Principle; Q216301	Hosting requirements -> qualification evidence
4.19	Installation and Deployment Specification	M	Defines repeatable installation/deployment, prerequisites, configuration, verification, rollback and record requirements.	Technical Lead	Infrastructure; Supplier; Validation	Peer	Q187219 4.7	Design baseline -> controlled deployment process
4.20	Final Design Review / Design Qualification Report	M	Confirms design completeness, consistency, feasibility, maintainability, traceability, regulatory compliance and acceptable risk before acceptance verification.	Validation Lead / Design Authority	Business; IT; QA; Supplier	Peer; QA	Q187219 4.5 and Appendix 1	Approved URS/design/risk -> approved design gate

5 - Build and Configuration

ID	Document / record	M/C	Purpose	Owner	Reviewer(s)	Approver(s)	Regulatory / SOP basis	Inputs -> outputs
5.01	Build and Configuration Plan	M	Defines build approach, environments, responsibilities, configuration controls, standards and evidence.	Configuration Lead	Validation; Supplier; QA	Project governance	Q187219 4.6	Approved design -> controlled build plan
5.02	Configuration Item Register and Initial Baseline	M	Lists application, database, middleware, OS, interfaces, reports, code, documents and critical configuration versions.	IT System Manager / Technical Lead	Infrastructure; Supplier; Validation	Peer	Q187219 4.7; Annex 11 10	Design -> baseline register
5.03	Environment Build Records	M	Records actual environment provisioning, versions, settings, accounts, network, monitoring, backup and deviations from design.	Infrastructure / Supplier	Technical Lead; Validation	Peer	Q187219 4.7	Installation spec -> as-built evidence
5.04	Application Configuration Records	M	Records configured values and links each item to approved CS and change/work item.	Configuration Lead	Business SME; Validation	Peer	Q187219 4.6; Annex 11 10	CS -> as-built configuration
5.05	Role and Permission Configuration Record	M	Records roles, privileges, data scope, approval rights, privileged accounts and service accounts.	IT Security Administrator	Business; QA; Security	Business Process Owner / IT	Annex 11 12; 21 CFR 11.10(g)	Role design -> configured access model
5.06	Master Data Build and Load Records	M	Records approved specifications, methods, units, limits, test definitions and other controlled master data loaded before verification.	Data Steward / Lab SME	Independent reviewer; QA	Data Owner / QA	Data integrity guidance; Annex 11 6/10	Approved master data -> controlled load evidence
5.07	Interface Build and Deployment Records	M	Records interface components, mappings, certificates/keys, schedules, error queues, monitoring and versions.	Interface Lead	Architect; Security; Validation	Peer	Annex 11 5; Q187219 4.6/4.7	Interface design -> as-built interface
5.08	Report and Certificate Configuration Records	M	Records report versions, layouts, data sources, formulas, signatures and deployment evidence.	Report Developer / Supplier	Business; QA; Validation	Peer	Annex 11 8	Report design -> as-built reports
5.09	Source Code and Version Control Records	C	Maintains traceable code, commits, tags, branches, reviews and releases for custom components.	Development Lead	Peer developer; Security	Development authority	Q187219 4.6; 21 CFR 11.10(k)	DS -> controlled code baseline
5.10	Code Review Report	C	Confirms custom code follows standards, design, security and maintainability expectations and is independently reviewed.	Development Lead	Independent qualified reviewer	Peer	Q187219 4.6; Q0300378 section 5	Code -> approved review findings
5.11	Development Verification Evidence	C	Records unit, integration and system tests performed during construction and defect resolution.	Development Lead / Supplier	Independent reviewer	Peer	Q187219 4.6	Code/configuration -> development evidence
5.12	Installation / Deployment Guide	M	Provides controlled steps for installation, deployment, configuration, rollback and validation evidence capture.	Technical Lead / Supplier	Infrastructure; Validation	Peer	Q187219 4.7; Q0300378 design	Design -> repeatable deployment method

ID	Document / record	M/C	Purpose	Owner	Reviewer(s)	Approver(s)	Regulatory / SOP basis	Inputs -> outputs
5.13	Test Data Management Plan	M	Defines representative, boundary, negative and production-like test data; masking; ownership; retention and clean-up.	Test Lead / Data Owner	QA; Security; Privacy	Validation Lead	Annex 11 4.7; data integrity principles	Verification scope -> controlled test data
5.14	Defect and Issue Log	M	Records classification, impact, root cause, correction, retest, disposition and release decision for defects.	Test Lead / Supplier	Business; Validation; QA	Per severity and QMS	Annex 11 4.2; Q187219 4.6/4.8	Build/testing -> controlled defect status
5.15	Supplier Release Notes and Component Evidence	M	Documents delivered version, fixed/known issues, dependencies, security status and supplier verification.	Supplier	IT System Manager; Validation; Security	Acceptance per VPL	Annex 11 3/4.5; Q187219 4.1	Supplier release -> accepted evidence
5.16	Draft SOPs, Work Instructions and Training Materials	M	Prepares operational controls and role-based training before release.	Process Owner / IT System Manager	End users; QA; Training	Process Owner; QA as required	21 CFR 11.10(i); Q187219 4.8/4.9	Design/risks -> operational procedures
5.17	Configuration Baseline Before Verification	M	Freezes and records the testable baseline before formal verification begins.	Technical Lead	Validation Lead; Supplier	Validation Lead	Q187219 4.7	As-built records -> verification baseline

17 lifecycle documents/records in this phase

6 - Testing and Qualification

ID	Document / record	M/C	Purpose	Owner	Reviewer(s)	Approver(s)	Regulatory / SOP basis	Inputs -> outputs
6.01	Acceptance Verification Plan	M	Defines IV/OV/PfV/pIV/pPfV strategy, environments, test coverage, acceptance criteria, evidence, deviations and traceability.	Validation Lead / Test Lead	Business; IT; Supplier	Peer; QA for plan/critical criteria	Q187219 4.8; Annex 11 4.1/4.7	Approved design/risk -> approved verification plan
6.02	Verification Environment Suitability Assessment	M	Confirms test environment similarity, control, segregation, configuration and data suitability.	Technical Lead / Validation	Infrastructure; Supplier	Validation Lead	Q187219 4.8	Environment baseline -> suitability conclusion
6.03	Installation Verification (IV/IQ) Protocol and Report	M	Verifies installation/deployment, versions, configuration, prerequisites, infrastructure references and documentation.	Test Lead	Independent reviewer	Peer	Q187219 4.8 Appendix 1	Baseline -> verified installation
6.04	Operational Verification (OV/OQ) Protocol and Report	M	Verifies functions throughout operating ranges, controls, errors, calculations, workflows and security.	Test Lead	Independent reviewer; SMEs	Peer	Q187219 4.8 Appendix 1	FS/CS/risk controls -> verified operation
6.05	Performance Verification (PfV/PQ/UAT) Protocol and Report	M	Demonstrates effective support of intended use and user requirements using representative business scenarios.	Business Test Lead	End users; Validation	Peer; QA	Q187219 4.8 Appendix 1	URS -> verified intended use
6.06	Production Installation Verification (pIV)	C	Verifies the production deployment and environment against approved design and release baseline.	Technical Lead	Independent reviewer	Peer	Q187219 4.8	Production deployment -> verified production installation
6.07	Production Performance Verification (pPfV)	C	Confirms selected critical workflows and interfaces perform effectively in production after deployment.	Business Test Lead	Validation; QA	Peer; QA	Q187219 4.8	Production state -> verified production performance

ID	Document / record	M/C	Purpose	Owner	Reviewer(s)	Approver(s)	Regulatory / SOP basis	Inputs -> outputs
6.08	Audit Trail Verification Protocol and Report	M	Challenges creation, security, content, reason capture, filtering, review, export and retention of audit trails.	Validation / QA	Business; IT; Supplier	QA	Annex 11 9; 21 CFR 11.10(e)	Audit trail design -> verified auditability
6.09	Electronic Signature Verification	C	Verifies identity, authentication, signature meaning, manifestation, linkage, uniqueness and invalid-use prevention.	Validation / QA	Business; Security; IT	QA	Annex 11 14; 21 CFR Part 11	E-signature design -> verified signatures
6.10	User Access and Segregation of Duties Verification	M	Verifies role permissions, approval authority, least privilege, conflicts, privileged/service accounts and access revocation.	Validation / Security	Business; QA	Business Process Owner; QA	Annex 11 12; 21 CFR 11.10(d)/(g)	Role design -> verified access model
6.11	Interface Verification and Reconciliation Report	M	Verifies mapping, completeness, accuracy, sequence, duplicates, retries, error handling, security and end-to-end reconciliation.	Interface Test Lead	Data Owner; Business; QA	Peer; QA	Annex 11 5	Interface design -> verified data exchange
6.12	Calculation and Scientific Logic Verification	M	Verifies formulas, units, rounding, limits, specifications, flags, invalid inputs and boundary conditions.	Lab SME / Test Lead	Independent scientific reviewer; QA	Business Process Owner; QA	Annex 11 4.7/6	Scientific requirements -> verified calculations
6.13	Report, Certificate and True Copy Verification	M	Verifies accurate/complete content, version, signatures, changes, export and human-readable/electronic copies.	Business Test Lead	Data Owner; QA	Peer; QA	Annex 11 8; 21 CFR 11.10(b)	Report design -> verified outputs
6.14	Backup, Restore and Recovery Verification	M	Verifies backup completeness, integrity, restore of application/configuration/data and achievement of RPO/RTO/WRT.	Infrastructure / Supplier	IT System Manager; Data Owner; Business	Peer	Annex 11 7.2/16; Q187219 4.11	Recovery design -> verified restore/recovery
6.15	Business Continuity / Manual Fallback Test	M	Exercises fallback activation, manual record controls, reconciliation and return to normal operation.	Business Continuity Owner	Lab Operations; IT; QA	Business Process Owner; QA	Annex 11 16	Continuity plan -> tested fallback
6.16	Security, Performance and Load Test Report	M	Verifies security controls, performance, capacity, concurrency, stress and monitoring thresholds based on risk.	Security / Performance Test Lead	Architect; Business; QA	Security/IT authority	Annex 11 4.7/12	NFRs -> verified resilience
6.17	Data Migration Verification and Reconciliation Report	C	Confirms complete and accurate migration, preserved meaning/metadata, exceptions, reconciliation and source disposition.	Migration Lead / Data Owner	Business; QA; IT	Data Owner; QA	Annex 11 4.8	Migration run -> accepted migrated data
6.18	Master Data Verification Report	M	Confirms approved master data is complete, accurate, effective-dated, versioned and linked to correct workflows.	Data Steward / Test Lead	Independent reviewer; QA	Data Owner; QA	Annex 11 6/10; data integrity guidance	Master data load -> verified baseline
6.19	Requirements and Design Traceability Matrix	M	Demonstrates traceability from business process/URS through risk, design, configuration, verification, deviations and release.	Validation Lead	Business; IT; QA	QA	Annex 11 4.4; Q187219 4.8	Lifecycle evidence -> coverage and gap view
6.20	Validation Deviations and Defect Assessments	M	Documents failed acceptance criteria, impact to critical aspects/GxP data, correction, retest and disposition.	Test Lead / Validation Lead	Business; IT; QA	QA per severity	Annex 11 4.2; Q187219 4.8	Test failures -> approved disposition

ID	Document / record	M/C	Purpose	Owner	Reviewer(s)	Approver(s)	Regulatory / SOP basis	Inputs -> outputs
6.21	Acceptance Verification Summary Report	M	Concludes verification success against acceptance criteria and readiness for deployment/release.	Validation Lead	Business; IT; Supplier	Peer; QA as applicable	Q187219 4.8	Executed verification -> release recommendation
6.22	Post-Verification Configuration Baseline	M	Records final tested configuration and versions before production release.	Technical Lead	Validation; IT Operations	Peer	Q187219 4.7	Verified system -> release baseline

22 lifecycle documents/records in this phase

7 - Release

ID	Document / record	M/C	Purpose	Owner	Reviewer(s)	Approver(s)	Regulatory / SOP basis	Inputs -> outputs
7.01	Release Readiness Checklist	M	Confirms validation, risks, procedures, training, access, support, data, supplier and infrastructure prerequisites are complete.	Validation Lead / IT System Manager	Business; IT; QA; Supplier	QA release gate	Q187219 4.8	Lifecycle evidence -> release decision inputs
7.02	Operational Readiness and Handover Checklist	M	Transfers ownership, support, monitoring, recurring tasks, documentation and known issues to operations.	Project Manager / IT System Manager	Operations; Business; Supplier	IT System Owner	Q187219 4.8	Implementation -> operational ownership
7.03	Approved Operation and Maintenance Description	M	Defines controlled operational activities, responsibilities, recurring reviews, changes, incidents, backup, supplier oversight and periodic evaluation.	IT System Manager	Business; Data Owner; IT SMEs	Peer signature; QA signature	Q187219 4.9 Appendix 1	Risks/design -> approved operating model
7.04	Approved SOPs and Work Instructions	M	Provides controlled procedures for users, administrators, data governance, review, changes, incidents and continuity.	Process Owners	SMEs; QA; Training	Per QMS	21 CFR 11.10(i)(j)/(k); Annex 11 operational sections	Draft SOPs -> effective procedures
7.05	Training Completion and Qualification Records	M	Demonstrates users and support personnel are trained and competent before access/use.	Training Coordinator / Managers	Business Process Owner; QA	Line management / QA as applicable	21 CFR 11.10(i); Q187219 4.8	Curriculum -> trained authorised users
7.06	Initial User Access Provisioning and Review Report	M	Confirms authorised users, roles, privileged/service accounts, approvals and removal of test accounts.	IT Access Manager	Line managers; Business; QA	Business Process Owner	Annex 11 12; Q187219 4.8/4.9	Access design/training -> production access baseline
7.07	Go-Live Cutover and Rollback Plan	M	Defines deployment sequence, data cutover, downtime, communications, checks, decision points and rollback.	Release Manager	Business; IT; Supplier; QA	Change authority / QA	Annex 11 10/16; Q187219 4.2/4.8	Release baseline -> controlled go-live
7.08	Go-Live / Cutover Execution Report	M	Records actual cutover, checks, issues, decisions, reconciliation and final status.	Release Manager	Business; IT; QA	Change authority	Q187219 4.8	Cutover execution -> production state evidence
7.09	Validation / Implementation Report (VPR)	M	Reports completion of planned activities, major changes, open items and conclusion that the validated state is established and LIMS is fit for intended use.	Validation Lead / IT System Manager	Business; Technical SMEs	IT System Owner signature; QA signature	Q187219 4.8 Appendix 1	All lifecycle evidence -> validation conclusion

ID	Document / record	M/C	Purpose	Owner	Reviewer(s)	Approver(s)	Regulatory / SOP basis	Inputs -> outputs
7.10	QA Release Approval	M	Formally releases LIMS for GxP use or confirms pre-approved release mechanism in the approved O&M description.	QA	Validation Lead; Business; IT	QA	Q187219 4.8	Readiness evidence -> authorised GxP use
7.11	Known Issues and Residual Risk Register	M	Documents unresolved items, impact, workarounds, owners, due dates, monitoring and release rationale.	IT System Manager / Risk Owners	Business; QA; Security	IT System Owner; QA where quality impact	ICH Q9(R1); Q187219 4.8	Open issues -> controlled post-release actions
7.12	Hypercare and Enhanced Monitoring Plan	C	Defines elevated support, data checks, interface monitoring, incident triage and exit criteria for early-life operation.	IT System Manager / Supplier	Business; QA	IT System Owner	Good IT practice; Q187219 4.9	Go-live risk -> stabilisation controls
7.13	Current Documentation Index	M	Provides inspector-ready list of document IDs, versions, approvals, locations and applicability.	Validation Lead / Document Control	IT System Manager; QA	QA/acceptance	Annex 11 4.1/4.3; Q187219 4.8	Approved package -> current document overview
7.14	Service Acceptance and Supplier Handover Record	M	Confirms service levels, support, escalation, operational reports, security evidence and supplier responsibilities are active.	IT System Manager	Procurement; QA; Business	IT System Owner	Annex 11 3; Q187219 4.1/4.8	Contract/O&M -> accepted service

14 lifecycle documents/records in this phase

8 - Operation and Maintenance

ID	Document / record	M/C	Purpose	Owner	Reviewer(s)	Approver(s)	Regulatory / SOP basis	Inputs -> outputs
8.01	Maintained O&M Description	M	Master operational control document covering roles, risks, supplier, changes, configuration, incidents, backup, access, monitoring and periodic evaluation.	IT System Manager	Business; Data Owner; QA	Peer; QA	Q187219 4.9	Validated state -> controlled operations
8.02	Maintained System Description and Configuration Baseline	M	Keeps architecture, data flows, interfaces, prerequisites, security and critical component versions current.	IT System Manager	Architect; Data Owner; QA	Per QMS	Annex 11 4.3/10	Operational changes -> current description/baseline
8.03	User and Access Management Procedure and Records	M	Controls authorisation, provisioning, modification, revocation, privileged/service accounts and approvals.	IT Access Manager / Line Managers	Business; Security; QA	Business Process Owner	Annex 11 12; 21 CFR 11.10(d)/(g)	Access requests -> current authorised access
8.04	Annual User Access Review Report	M	Confirms access remains aligned to job roles, least privilege and segregation of duties across full technology stack.	Line Managers / IT Access Manager	Business; Security; QA	Business Process Owner / delegated approver	Q187219 4.9; Q0300378 7.2	User/access lists -> approved corrections
8.05	Audit Trail Review Procedure and Records	M	Defines risk-based event scope, frequency, reviewer independence, evidence, escalation and trend analysis.	Business Process Owner / QA	Data Owner; IT	QA / Process Owner	Annex 11 9; FDA DI guidance	Audit trail data -> documented review and actions
8.06	GxP Data Review and Approval Procedure	M	Defines review of results, metadata, calculations, exceptions, attachments, signatures and audit trail before approval/use.	Lab Process Owner	QA; Data Owner	Business Process Owner / QA as required	Data integrity guidance; Annex 11 6/9	LIMS records -> approved reliable data

ID	Document / record	M/C	Purpose	Owner	Reviewer(s)	Approver(s)	Regulatory / SOP basis	Inputs -> outputs
8.07	Master Data Management Procedure and Change Records	M	Controls creation, review, approval, effective dating, versioning, deployment and periodic review of master data.	Data Owner / Data Stewards	Lab SMEs; QA; IT	Data Owner; QA	Annex 11 10; PIC/S PI 041-1	Master data requests -> approved effective records
8.08	Interface Monitoring and Reconciliation Procedure / Logs	M	Monitors transactions, errors, missing/duplicate records, retries, reconciliation and escalation.	Interface Service Owner	Data Owner; Business; QA	IT System Manager / Data Owner	Annex 11 5	Interface messages -> reconciled data flow
8.09	IT Change Records	M	Evaluates, authorises, implements, verifies and closes normal, standard and emergency changes.	Change Owner	Business; IT; QA	Change authority / QA as applicable	Annex 11 10; Q187219 4.2	Change request -> controlled implemented change
8.10	Configuration Management Records	M	Maintains versions and status of application, infrastructure, interfaces, code, reports, documents and critical settings.	IT System Manager	Technical SMEs; Supplier	Peer / change authority	Annex 11 10; Q187219 4.6/4.7	Approved changes -> updated baseline
8.11	IT Incident and Problem Records	M	Records impact, triage, workaround, root cause, correction, CAPA, communication and trend for operational events.	Service Desk / IT System Manager	Business; QA; Security	Per incident/deviation process	Annex 11 13; Q187219 4.10	Events -> restored service and prevention
8.12	Quality Deviation and CAPA Records	C	Manages events affecting product quality, patient safety, validated state, regulatory records or GxP data.	Business / QA	IT; Data Owner; Supplier	QA	Annex 11 13; internal deviation process	Quality event -> approved investigation/CAPA
8.13	Backup Monitoring and Restore Test Records	M	Monitors backup success/integrity and periodically demonstrates accurate, complete restore and recovery.	Infrastructure / Supplier	IT System Manager; Data Owner; QA	Peer	Annex 11 7.2; Q187219 4.11	Backup jobs -> assurance of recoverability
8.14	IT Recovery Plan and Exercise Records	M	Maintains recovery procedures, dependencies, contacts, targets and exercise evidence.	IT System Manager / Infrastructure	Business Continuity; Supplier; QA	IT System Owner	Annex 11 16	Recovery design -> tested operational recovery
8.15	Business Continuity / Manual Fallback Procedure and Tests	M	Maintains alternate processes, controlled manual records, reconciliation and periodic exercises.	Business Process Owner	Lab Operations; IT; QA	Business Process Owner; QA	Annex 11 16	Fallback requirements -> operational continuity
8.16	Security, Vulnerability and Patch Management Records	M	Documents threat monitoring, vulnerability assessment, patch decisions, compensating controls, logs and security incidents.	Security / IT Operations	IT System Manager; Supplier; QA	Security authority / change authority	Annex 11 12; Q187655 reference	Security inputs -> maintained risk control
8.17	Supplier Performance Monitoring and Service Review	M	Reviews SLAs/KPIs, incidents, changes, security evidence, release notices, subcontractors, continuity and action items.	IT System Manager / Supplier Manager	Business; QA; Security; Procurement	IT System Owner	Annex 11 3; Q187219 4.1	Supplier reports -> accepted performance/actions
8.18	Supplier Annual Re-attestation / Reassessment	M	Confirms supplier assessment remains valid and triggers reassessment when material changes occur.	Supplier Manager	QA; Security; Procurement	Peer; QA quality aspects	Q187219 4.1	Monitoring/change -> current supplier assurance
8.19	Software/Hardware Lifecycle and Obsolescence Plan	M	Tracks support status, end-of-life, technical debt, upgrade roadmaps and risk mitigations.	IT System Manager / Architect	Supplier; Security; Business; QA	IT System Owner	Annex 11 11; Q187219 4.9	Vendor/CMDB data -> lifecycle actions
8.20	Training Curriculum and Ongoing Training Records	M	Maintains role-based initial and refresher training and verifies competence after significant change.	Business Process Owner / Managers	Training; QA; IT	Line management / QA as applicable	21 CFR 11.10(i)	SOP/change -> competent users

ID	Document / record	M/C	Purpose	Owner	Reviewer(s)	Approver(s)	Regulatory / SOP basis	Inputs -> outputs
8.21	Data Quality and Integrity Monitoring Dashboard	M	Trends invalid results, corrections, audit trail events, overrides, retests, interface errors, late entries and review timeliness.	Data Owner / QA	Business; IT	Data Governance forum	ICH Q10; FDA DI guidance	Operational records -> risk-based oversight
8.22	Backend Data Change Request and Execution Record	C	Controls exceptional direct data changes with approval, technical execution, before/after evidence, reconciliation and auditability.	Data Owner / IT	QA; Business; Supplier	QA / authorised owner	21 CFR 11.10(e); data integrity guidance	Approved exception -> traceable corrected data
8.23	Archive, Retrieval and Legal Hold Records	M	Documents archival transfers, retrieval tests, legal holds, access, readability and integrity over retention period.	Records Management / Data Owner	QA; IT	Data Owner / QA	Annex 11 17; 21 CFR 11.10(c)	Records -> retrievable retained evidence
8.24	Data Extract, True Copy and Inspection Request Records	M	Controls generation, verification, transfer and chain-of-custody of inspection/review copies.	Data Owner / IT System Manager	QA; Legal; Business	Data Owner / QA	Annex 11 8; 21 CFR 11.10(b)/(c)	Request -> verified complete copy
8.25	Operational Metrics and Management Review Records	M	Reviews availability, performance, incidents, problems, changes, risks, data quality, supplier and compliance performance.	IT System Manager / Business Process Owner	QA; Data Owner; Supplier	IT System Owner / management	ICH Q10; Q187219 4.9	Operational metrics -> decisions/actions

25 lifecycle documents/records in this phase

9 - Periodic Evaluation

ID	Document / record	M/C	Purpose	Owner	Reviewer(s)	Approver(s)	Regulatory / SOP basis	Inputs -> outputs
9.01	Periodic System Evaluation (IT PSE) Plan and Report	M	Concludes whether LIMS remains validated, fit for intended use and compliant with health-authority requirements.	IT System Manager / Business Process Owner	Data Owner; Security; Supplier	IT System Owner signature; QA signature	Annex 11 11; Q187219 4.9 Appendix 1	Periodic evidence -> validated-state conclusion
9.02	Current Functionality and Intended Use Review	M	Confirms actual use, sites, users, records, interfaces and functions remain within approved intended use.	Business Process Owner	IT; QA; Data Owner	Business Process Owner; QA	Annex 11 11	System description/usage -> scope conclusion
9.03	ITRA Review and Re-attestation	M	Confirms risk assumptions, controls and residual risks remain current and effective.	IT System Manager	QA; Security; Data Owner; Business	IT System Owner; QA	Q187219 section 2/4.9	Risk/control evidence -> updated ITRA
9.04	Functional and Data Integrity Risk Review	M	Reassesses critical aspects and data lifecycle risks based on incidents, changes, trends and new knowledge.	Validation Lead / Data Owner	Business; QA; IT	QA / Data Owner	ICH Q9(R1); FDA DI guidance	Operational experience -> current risk profile
9.05	User Access Review Summary	M	Confirms annual access review completion, privileged account status and closure of findings.	Business Process Owner / IT	QA; Security	Business Process Owner	Annex 11 12; Q187219 4.9	Access review -> periodic assurance
9.06	Audit Trail Review Effectiveness Assessment	M	Evaluates review coverage, timeliness, findings, trends, risk-based frequency and need for procedural/system improvements.	QA / Business Process Owner	Data Owner; IT	QA	Annex 11 9/11	Audit trail reviews -> effectiveness conclusion
9.07	Incident, Problem, Deviation and CAPA Trend Review	M	Assesses recurrence, severity, root causes, effectiveness and impact on validated state.	IT System Manager / QA	Business; Supplier	QA / management	Annex 11 11/13	Operational records -> trend/CAPA actions

ID	Document / record	M/C	Purpose	Owner	Reviewer(s)	Approver(s)	Regulatory / SOP basis	Inputs -> outputs
9.08	Change and Release History Review	M	Assesses cumulative change, verification adequacy, regression coverage and documentation/configuration currency.	IT System Manager / Validation	QA; Business; Supplier	QA	Annex 11 10/11	Change history -> validated-state assurance
9.09	Backup, Restore, Recovery and Continuity Review	M	Confirms monitoring, test results, targets, dependencies and corrective actions remain adequate.	IT System Manager / Business Continuity	Data Owner; QA; Supplier	IT System Owner	Annex 11 7.2/11/16	Recovery evidence -> continuity assurance
9.10	Security, Patch and Vulnerability Review	M	Assesses security events, vulnerabilities, patches, hardening, access attempts and emerging threats.	Security / IT System Manager	Supplier; QA	Security authority / IT System Owner	Annex 11 11/12	Security evidence -> risk actions
9.11	Supplier and SLA/KPI Review	M	Confirms supplier performance, assessment validity, audit status, changes, subcontractors, continuity and open actions.	Supplier Manager	QA; Procurement; Business; Security	IT System Owner	Annex 11 3/11	Supplier evidence -> continued suitability
9.12	Configuration, Support and Obsolescence Review	M	Confirms supported versions, current baseline, critical components, licenses and lifecycle plans.	IT System Manager / Architect	Supplier; QA	IT System Owner	Annex 11 10/11	CMDB/vendor data -> lifecycle decision
9.13	Data Integrity and ALCOA+ Health Check	M	Assesses trends in corrections, invalidations, retests, late entries, overrides, audit trail findings, interface errors and review effectiveness.	Data Owner / QA	Business; IT	Data Governance forum / QA	FDA DI guidance; PIC/S PI 041-1	Data metrics -> DI conclusion/actions
9.14	Documentation, SOP and Training Currency Review	M	Confirms controlled documents, current index, training curricula and completion remain aligned to system use.	Document Owner / Training	IT; Business; QA	QA / owners	21 CFR 11.10(i)(k); Annex 11 11	Document/training records -> currency conclusion
9.15	Periodic Evaluation Action and CAPA Plan	M	Assigns owners, priorities, due dates and escalation for identified gaps and improvements.	IT System Manager / QA	Risk owners; management	QA / IT System Owner	ICH Q10; Annex 11 11	PSE findings -> controlled actions

15 lifecycle documents/records in this phase

10 - Retirement

ID	Document / record	M/C	Purpose	Owner	Reviewer(s)	Approver(s)	Regulatory / SOP basis	Inputs -> outputs
10.01	Retirement Change Request	M	Authorises retirement and evaluates impact on processes, records, interfaces, infrastructure, contracts and validated state.	Change Owner	Business; IT; Data Owner; QA	IT System Owner; QA	Annex 11 10; Q187219 4.12	Retirement decision -> approved change
10.02	Retirement Plan	M	Defines system/data disposition, migration/archive, verification, access, interfaces, contracts, communication, responsibilities and timeline.	IT System Manager / Business Process Owner	Data Owner; Records; Legal; Security; Supplier	IT System Owner; QA	Q187219 4.12 Appendix 1; Annex 11 17	Approved change -> retirement strategy

ID	Document / record	M/C	Purpose	Owner	Reviewer(s)	Approver(s)	Regulatory / SOP basis	Inputs -> outputs
10.03	Final Periodic Evaluation	M	Documents that LIMS remained in a validated state and under control until retirement.	IT System Manager / Business Process Owner	Data Owner; QA	IT System Owner; QA	Q187219 4.12	Operational evidence -> final validated-state conclusion
10.04	Data Disposition and Retention Assessment	M	Identifies every GxP record, metadata, audit trail, signature, attachment and configuration item and assigns retention/disposition.	Data Owner / Records Management	Business; QA; Legal; IT	Data Owner; QA	Annex 11 7/17; 21 CFR 11.10(c)	Data inventory -> approved disposition
10.05	Archive / Migration Plan and Mapping	M	Defines extraction, format, metadata, mapping, transformation, reconciliation, access, security and rollback for retained data.	Data Owner / Migration Lead	QA; Business; IT; Supplier	Data Owner; QA	Annex 11 4.8/17	Disposition assessment -> archive/migration method
10.06	Data Extraction and Reconciliation Report	M	Demonstrates complete and accurate extraction/transfer of records and metadata with exceptions resolved.	Migration Lead	Independent reviewer; QA; Business	Data Owner; QA	Annex 11 4.8	Executed extraction -> accepted data set
10.07	Archive Verification and Retrieval Test Report	M	Confirms retained data is accessible, readable, complete, accurate, intact and retrievable within required time.	Records Management / Data Owner	QA; IT	Data Owner; QA	Annex 11 17; 21 CFR 11.10(c)	Archive -> verified long-term access
10.08	User and Privileged Access Decommission Record	M	Removes or restricts access while preserving authorised archive/inspection access.	IT Access Manager	Security; Data Owner; QA	IT System Manager	Annex 11 12/17	Retirement plan -> controlled access closure
10.09	Interface and Dependency Decommission Record	M	Confirms downstream/upstream dependencies are migrated, disabled and monitored without data loss.	Architect / Interface Lead	Business; Data Owner; QA	IT System Owner	Annex 11 5/16	Dependency map -> safe decommission
10.10	Infrastructure and Configuration Item Retirement Record	M	Updates lifecycle status, removes workloads, securely disposes media and preserves necessary evidence.	Infrastructure Owner / IT System Manager	Security; QA; Supplier	Infrastructure Owner / QA as applicable	Q216301 section 11; Q187219 4.12	Retirement plan -> retired CIs
10.11	Supplier Contract, License and Exit Closure Record	M	Closes services while preserving data access, documentation, audit rights and retention obligations.	Supplier Manager / Procurement	Legal; Data Owner; QA; IT	IT System Owner / Procurement	Annex 11 3/17	Contract/exit plan -> closed supplier obligations
10.12	Secure Data and Media Destruction Certificate	C	Provides approved and verifiable destruction evidence for data/media whose retention period has ended.	Data Owner / Security	Records Management; QA	Data Owner; QA	Annex 11 17; information-handling requirements	Disposition approval -> destruction evidence
10.13	ServiceNow and Repository Lifecycle Status Update	M	Updates business application, configuration items, access, documentation repositories and recurring tasks to reflect retirement.	IT System Manager	Document Control; Infrastructure	Per supporting tools	Q187219 4.12	Retirement execution -> accurate inventories
10.14	Retirement Report	M	Confirms planned activities, data disposition, control until retirement, open responsibilities and final conclusion.	IT System Manager / Business Process Owner	Data Owner; Legal; Records	IT System Owner; QA	Q187219 4.12 Appendix 1	Retirement evidence -> approved closure
10.15	Post-Retirement Inspection Support and Ownership Handover	M	Assigns responsibility, access method, contacts and evidence for authority requests throughout the retention period.	Data Owner / Records Management	QA; Legal; IT	Data Owner; QA	Q187219 4.12; Annex 11 17	Retirement report -> sustained inspection readiness

